BISHWA RANJAN SI

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and Institute for NanoBioTechnology (INBT)
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EDUCATION

Ph.D., Chemical & Biomolecular Engineering, Johns Hopkins University, MD 2021 - Expected 12/2026

B.S. and M.S., Chemical Engineering, Indian Institute of Technology Kanpur, India 2014 - 2019

RESEARCH EXPERIENCE

Johns Hopkins University, Chemical & Biomolecular Engineering and INBT 11/2021 - present
Graduate Research Assistant | Advisor: Konstantinos Konstantopoulos, Ph.D.

- Investigated how extracellular viscosity, confinement, matrix stiffness, and electric fields regulate cancer cell migration; demonstrated enhanced 3D invasion through KCC4-dependent signaling at elevated viscosity.
- Formulated tunable polyacrylamide hydrogels and characterized their rheology to relate healthy and tumor-associated breast tissue stiffness to cell behavior.
- Optimized fixation, permeabilization, immunolabeling, and confocal acquisition for quantitative imaging of 2D cultures and 3D samples. Mapped NHE1, KCC4, SWELL1, and cytoskeletal regulators using localization, line-scan, and polarization analyses.
- Engineered optogenetic constructs for targeted protein activation, recruitment, and degradation, with controlled expression and illumination for live-cell imaging.
- Standardized assays across 2D cultures, microfluidic devices, collagen-embedded spheroids, and zebrafish xenografts, including tumor-cell injection at three zebrafish sites. Established quantitative measures of localization, tracking, velocity, direction reversal, and 3D dissemination.
- Designed a microfluidic electrotaxis device to investigate confined cancer cell migration, contractility, and calcium signaling; work presented at conferences and developed into a manuscript in preparation.
- First or co-first author on studies in *Science Advances* (accepted) and *Cell Research*; co-author in *Nature Cell Biology* (accepted), *Oncogene*, and *Cancer Research*.

Indian Institute of Technology Kanpur, India 06/2020 - 05/2021
Project Associate | Advisor: Rahul Mangal, Ph.D.

- Investigated active liquid-crystal droplets and Janus colloids in complex fluids using rheology and optical microscopy; first-authored an Elsevier book chapter.

Indian Institute of Technology Kanpur, India 04/2018 - 06/2019
M.S. Research Assistant | Advisor: Rahul Mangal, Ph.D.

- Studied hydrodynamic and interfacial effects on active Janus colloids and liquid-crystal droplets using rheology, optical microscopy, SEM, and thin-film coating; contributed to papers in *Langmuir* and *Physics of Fluids*.

INDUSTRY EXPERIENCE

Sterlite Technologies, Pune, India 07/2019 - 04/2020
Post-graduate Engineer

- Completed three months of training in optical fiber manufacturing and cabling plants; identified process improvements to reduce downtime and enhance fiber drawing plant efficiency.

HONORS AND AWARDS

2023 CDI Travel Award, Chemical & Biomolecular Engineering, *Johns Hopkins University*

2021 Doctoral Fellowship, Chemical & Biomolecular Engineering, *Johns Hopkins University*

2018-19 MHRD Postgraduate Scholarship, Ministry of Human Resource Development, Government of India, *IIT Kanpur*

PUBLICATIONS

8 peer-reviewed publications and 2 accepted manuscripts; ***** indicates first or co-first authorship.

Accepted

1. **Si BR***, Choudhury D*, Bera K, Li Y, Stoletov K, Graziano G, Ceisel A, Gao J, Mehta P, Bustillo B, Tran A, Sen A, Hauf M, Ewald AJ, Lewis JD, Mumm JS, Sun SX, Konstantopoulos K. Extracellular fluid viscosity regulates cancer cell invasion via KCC4-dependent signaling. ***Science Advances***. **Accepted, 2026**.
2. Amitrano A*, Choudhury D*, Ifemembi B, Afthinos A, Stoletov K, Yuan Q, Nath S, **Si BR**, Agarwal B, Graziano G, Gao J, Ceisel A, Hauf M, Sun SX, Ewald AJ, Valverde MA, Lewis JD, Mumm JS, Konstantopoulos K. Mechanochemical cues control the coupling of metabolic and migratory patterns in cancer. ***Nature Cell Biology***. **Accepted, 2026**. Preprint: bioRxiv 2026.08.21.746296. doi:10.64898/2026.08.21.746296.

Peer-Reviewed Research Articles

3. Stoletov K, Karimian SS, Brown D, Amitrano A, **Si BR**, Verma VM, Gupta S, Bukhari AB, Lamb S, Konstantopoulos K, Wong GK, Lewis JD. Adrenomedullin 2 drives intratumoral cancer cell invasion, intravasation and metastasis. ***Oncogene*. 2026** Nov;45(45):4916-4926. doi:10.1038/s41388-026-04012-3. PMID: 42859189; PMCID: PMC13731145.
4. Lee G, Wong C, Cho A, West JJ, Crawford AJ, Russo GC, **Si BR**, Kim J, Hoffner L, Jang C, Jung M, Leone RD, Konstantopoulos K, Ewald AJ, Wirtz D, Jeong S. E-cadherin induces serine synthesis to support progression and metastasis of breast cancer. ***Cancer Research*. 2024** Sep;84(17):2820-2835. doi:10.1158/0008-5472.CAN-23-3082. PMID: 38959339; PMCID: PMC11374473.
5. Dwivedi P, **Si BR**, Pillai D, Mangal R. Solute induced jittery motion of self-propelled droplets. ***Physics of Fluids*. 2021** Feb;33(2):022103. doi:10.1063/5.0038716.
6. **Si BR**, Patel P, Mangal R. Self-propelled Janus colloids in shear flow. ***Langmuir*. 2020** Oct;36(40):11888-11898. doi:10.1021/acs.langmuir.0c01924. PMID: 32897720.

Methods, Reviews, Book Chapters, and Research Highlights

7. **Si BR***, Choudhury D*, Nath S*, Konstantopoulos K. Galvanin: a molecular compass for bioelectric navigation. ***Cell Research*. 2026** Nov;36(11):837-838. doi:10.1038/s41422-026-01279-9. PMID: 42409928. (Research Highlight)
8. **Si BR**, Yuan Q, Nath S, Agarwal B, Amitrano A, Choudhury D, Konstantopoulos K. Engineered in vitro platforms for mechanochemical control of cell migration. ***Cold Spring Harbor Perspectives in Biology*. 2025** Oct;17(10):a041795. doi:10.1101/cshperspect.a041795. PMID: 40588325; PMCID: PMC12487712. (Book Chapter)
9. Mistriotis P, Wisniewski EO, **Si BR**, Kalab P, Konstantopoulos K. Coordinated in confined migration: crosstalk between the nucleus and ion channel-mediated mechanosensation. ***Trends in Cell Biology*. 2024** Oct;34(10):809-825. doi:10.1016/j.tcb.2024.01.001. PMID: 38290913; PMCID: PMC11284253. (Review)
10. **Si BR**, Mangal R. Morphology of nanoparticle-based polymer composites. In: *Nanoparticle-Based Polymer Composites*. Elsevier; **2022**. pp. 151-179. doi:10.1016/B978-0-12-824272-8.00006-3. (Book Chapter)

Preprints and Manuscripts Under Review

11. Ge Z, Ni Q, Yan Y, Wu Y, **Si BR**, Fu J, Lin D, Konstantopoulos K, Li Y, Sun SX. Anionic lipid trafficking and cell mechanics regulate membrane electrical potential in non-excitable tissue cells. ***bioRxiv*. 2025**. doi:10.1101/2025.11.25.690536. Under review at ***Nature Materials***.
12. Sen A*, **Si BR***, Choudhury D*, Stoletov K, Agarwal B, Srivastava N, Nath S, Remmert J, Bera K, Ceisel A, Graziano G, Gao J, Wimalasena V, Hauf M, Lewis JD, Ewald AJ, Mumm JS, Konstantopoulos K. Viscous memory regulates cancer cell invasion in vitro and in vivo. **Submitted**.
13. Amitrano A*, Agarwal B*, Choudhury D, **Si BR**, Sloan I, Konstantopoulos K. Rethinking tumor acidosis: driver, brake, or adaptive challenge? ***American Journal of Physiology - Cell Physiology***. **Invited review**.

Manuscripts in Preparation

- Electric field-guided migration of confined cancer cells: roles of cellular contractility and calcium signaling.

PRESENTATIONS

Oral Presentations

2026 Elevated extracellular viscosity rewires polarized ion channel transport to drive confined cell migration. *AIChE* Annual Meeting, Minneapolis, MN.

2026 Elevated fluid viscosity rewires cancer cells to KCC4-dependent invasion. *BMES* Annual Meeting, Orlando, FL.

2025 Ion channel-mediated regulation of confined cell migration under elevated extracellular viscosity. *AIChE* Annual Meeting, Boston, MA.

2023 A novel microfluidic device to study the influence of electric field on cancer cell motility under confinement and its underlying effect on cellular contractility. *AIChE* Annual Meeting, Orlando, FL.

Poster Presentations

2026 Elevated fluid viscosity rewires cancer cells to KCC4-dependent invasion. *INBT Symposium*, Johns Hopkins University, Baltimore, MD.

2026 Elevated fluid viscosity rewires cancer cells to KCC4-dependent invasion. *DOM-WSE Research Retreat*, Department of Medicine and Whiting School of Engineering, Johns Hopkins University, Baltimore, MD.

2025 Extracellular fluid viscosity regulates distinct ion channel dynamics to enhance 3D cancer cell migration. *BMES* Annual Meeting, San Diego, CA.

2025 Electric field-guided migration of confined cancer cells: roles of cellular contractility and calcium signaling. *INBT Symposium*, Johns Hopkins University, Baltimore, MD.

2024 A microfluidic model for electrotaxis in tumor microenvironments: phenotypic modulations and calcium signaling. *BMES* Annual Meeting, Baltimore, MD.

RESEARCH SUPPORT

Current

NIH/NCI R01 CA257647 (PI: K. Konstantopoulos) 04/2021 - 03/2027
Viscotaxis: Novel Cell Migration Mechanisms Regulated by Microenvironmental Viscosity

Role: Graduate Research Assistant

Grant Development

- Contributed experimental data and grant writing to one funded NIH R01 and three additional NIH R01 applications, supporting studies of cancer cell behavior and therapeutic target discovery (PI: K. Konstantopoulos).

TEACHING EXPERIENCE

Graduate Teaching Assistant, Chemical & Biomolecular Engineering, Johns Hopkins University 2022 - 2023
EN.540.652, Advanced Transport Phenomena (Fall 2022 and Fall 2023). Delivered two lectures, led recitation and exam-review sessions, held weekly office hours, wrote problem sets and solutions, and graded homework and exams.

Teaching Assistant, Chemical Engineering, Indian Institute of Technology Kanpur 2018 - 2019
CHE200, Chemical Engineering Communication Skills (Fall 2018); CHE677, Introduction to Polymer Physics (Spring 2019).

RESEARCH MENTORING

Graduate Students, Johns Hopkins University (Konstantopoulos Lab)

- Jake Remmert (Ph.D. student); *Project*: Mechanical cues governing senolytic resistance and fibrosis.

- Bhawana Agarwal (Ph.D. student); *Project*: Viscous regulation of macrophage polarization and cancer-immune crosstalk; Paeonia Foundation travel award winner.
- Sanjiban Nath (Ph.D. student); *Project*: Stiffness-dependent traction in confinement.
- Nitish Srivastava (Ph.D. student); *Project*: Hypoxia and viscosity regulating cancer cell migration mode.
- Xuefei Lu (Ph.D. student); *Project*: Endocytosis regulating cancer cell migration dynamics.

Undergraduate Students

- Beatrice Bustillo (B.S. student, Johns Hopkins University); *Project*: Extracellular viscosity and KCC4-dependent cancer invasion; co-author in *Science Advances*; received a travel award for a summer research internship at EPFL.
- Preet Patel (B.S. student, IIT Kanpur); now graduate student at MIT Media Lab; *Project*: Self-propelled Janus colloids in shear flow; co-author in *Langmuir*.

PROFESSIONAL SERVICE & DEVELOPMENT

2026 Judge, Undergraduate Student Poster Competition, *AIChE* Annual Meeting, Minneapolis, MN

2015-17 Academic Mentor for junior undergraduates, Institute Counselling Service (now Center for Mental Health and Wellbeing), *IIT Kanpur*

TECHNICAL EXPERTISE

Hydrogels and Polymers: Polyacrylamide hydrogel formulation (tunable stiffness), collagen gel embedding, gel rheology, polymer-nanoparticle composites, thin-film coating, microfluidic fabrication.

Sample Preparation: Mammalian cell culture, 3D tumor spheroids, immunofluorescence, zebrafish xenograft microinjection and imaging.

Microscopy: Confocal and live-cell imaging, fluorescence lifetime imaging (FLIM), traction force microscopy, optogenetics, SEM, optical microscopy, acquisition optimization.

Image and Data Analysis: Python, MATLAB, Fiji/ImageJ, IMARIS, GraphPad Prism; segmentation, cell tracking, line-scan, polarization, and reproducibility analysis.

Molecular Biology: Lentiviral transduction, plasmid design, RNAi, CRISPR, RT-qPCR, Western blot, flow cytometry.

Metabolism Assays: Seahorse analysis, mitochondrial isolation/transplantation, metabolic biosensors.

REFERENCES

Konstantinos Konstantopoulos, Ph.D. (doctoral advisor)
Professor of Chemical & Biomolecular Engineering, Johns Hopkins University
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Sean X. Sun, Ph.D. (collaborator)
Professor of Mechanical Engineering, Johns Hopkins University
ssun@jhu.edu
Jeff Mumm, Ph.D. (collaborator)
Professor of Ophthalmology, Wilmer Eye Institute, Johns Hopkins University School of Medicine
jmumm3@jhmi.edu
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Assistant Professor of Chemical & Biomolecular Engineering, Johns Hopkins University
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